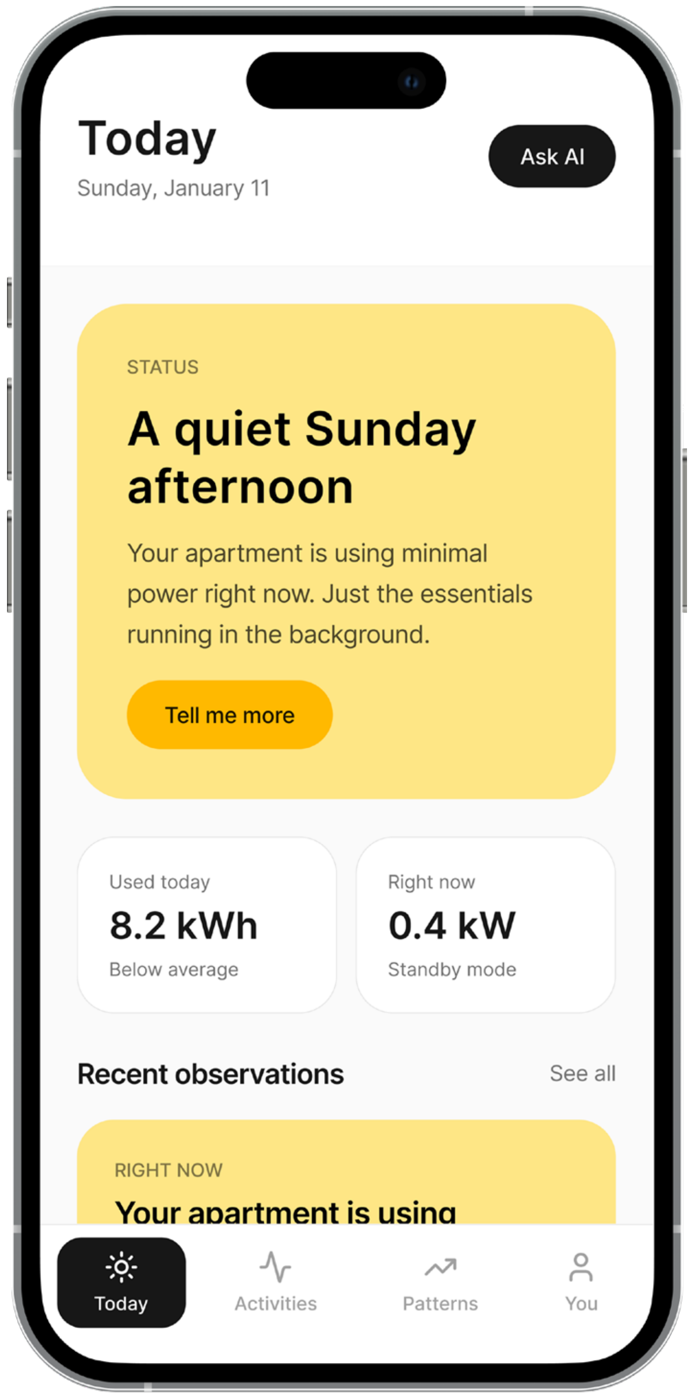
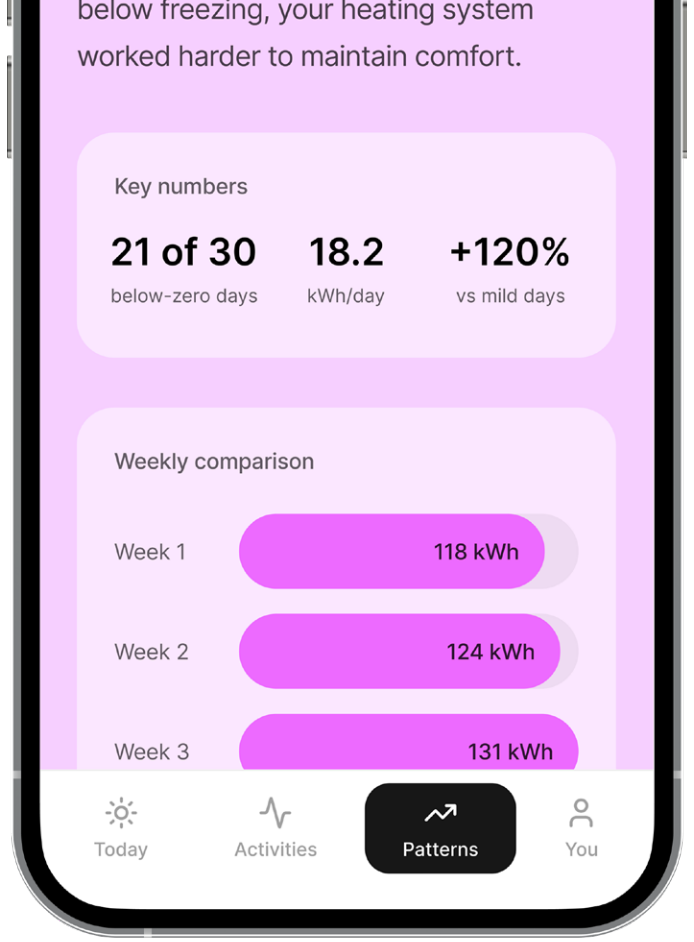
Aalto University School of Science and Technology

EnergyLM

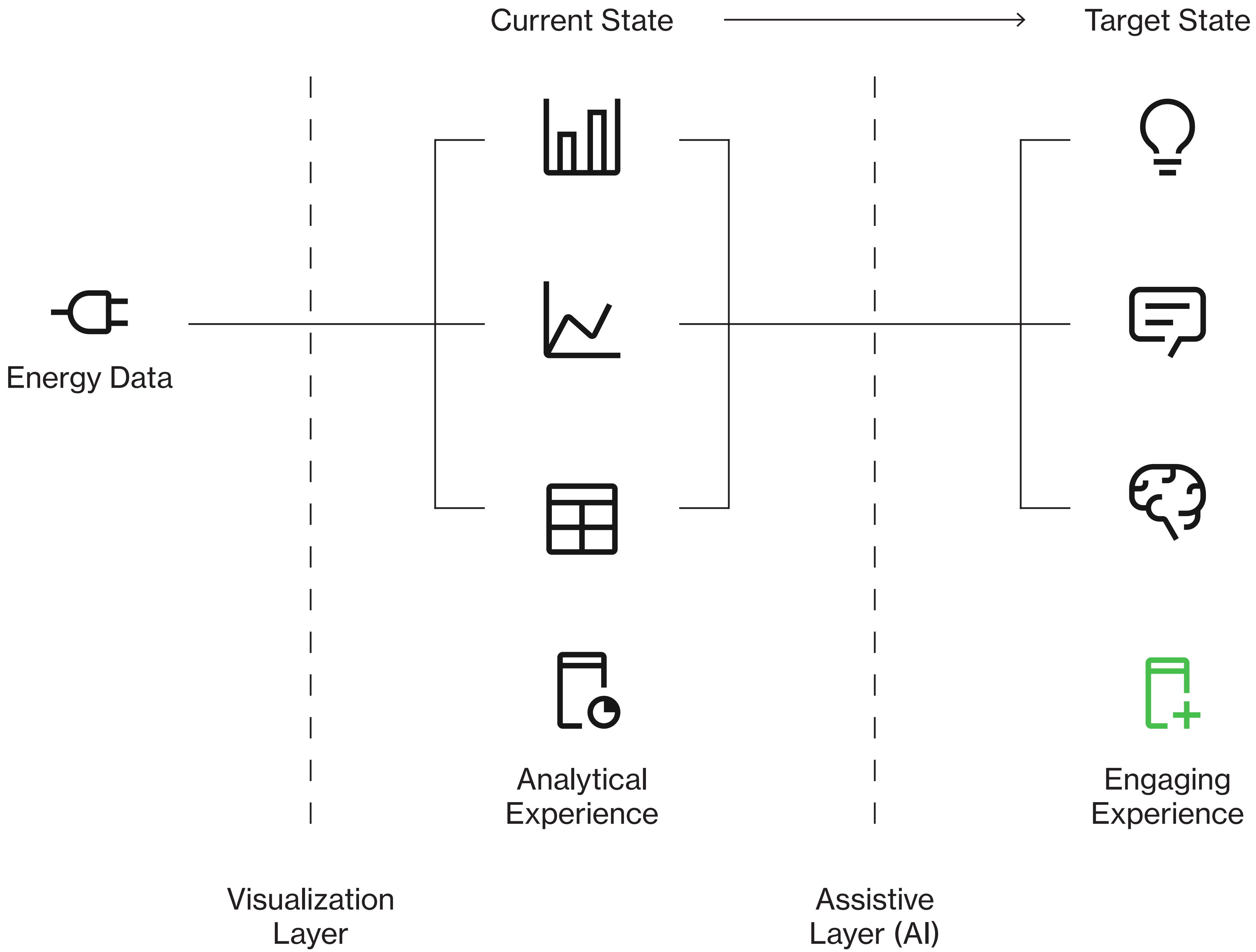
EnergyLM is an AI-mediated energy awareness app that explores how domestic energy consumption can be reframed through interpretation and situated understanding.

Developed within the course Advanced Topics in Human–Computer Interaction at Aalto University, the project investigates how AI can support energy awareness without automating or controlling user behavior, focusing instead on understanding, reflection, and informed choice.





Team project · 4 months



Home energy consumption apps today largely rely on quantitative monitoring, presenting users with charts, totals, and historical data. While technically accurate, these interfaces can struggle to support meaningful understanding or long-term engagement.

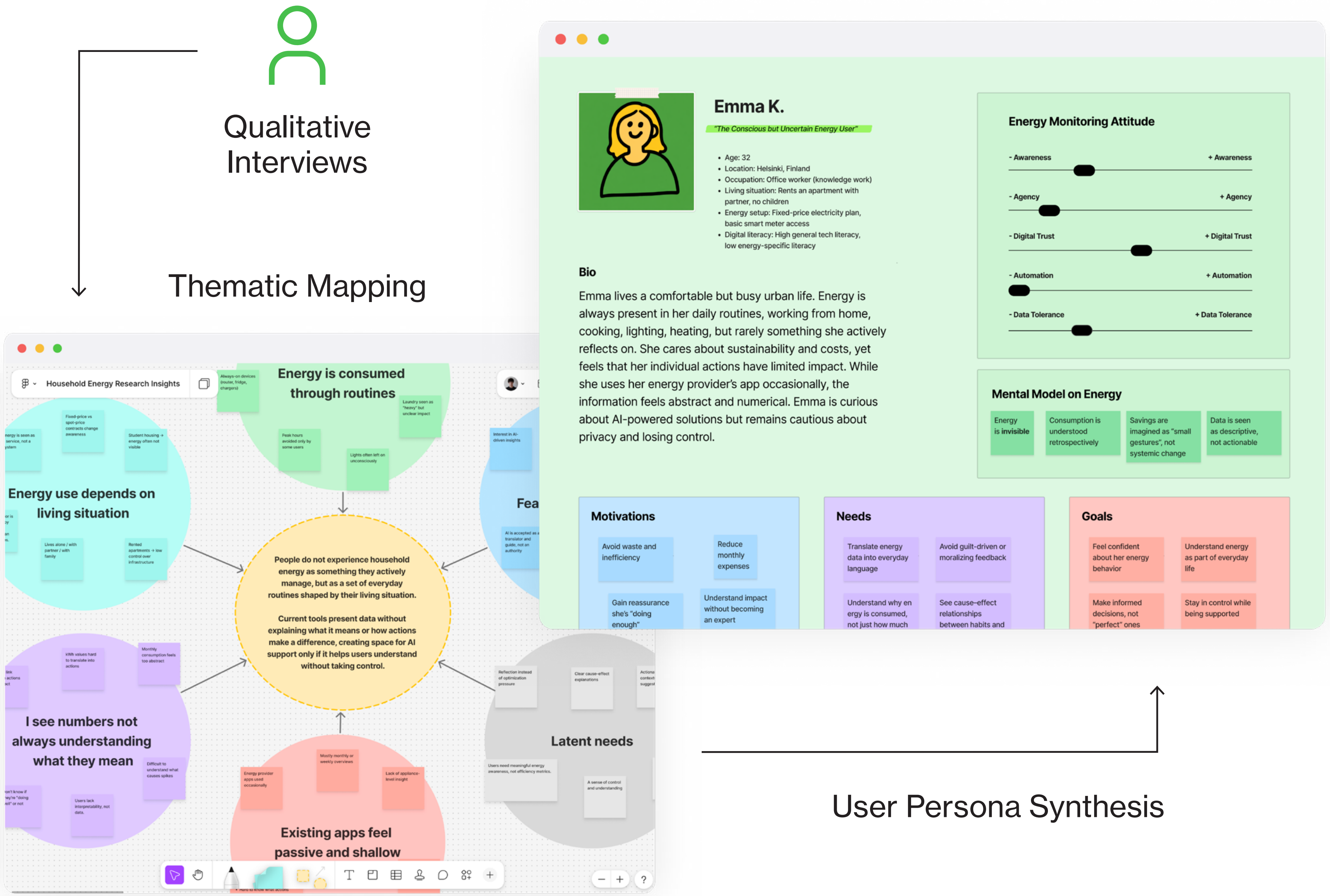
The brief of the course was therefore to design a new energy consumption app experience and explore possible ways of integrating AI.

Rather than assuming a predefined role for AI, our team approached the brief as an open question: how might AI be introduced into energy consumption apps in a way that is both effective and natural within everyday domestic contexts?

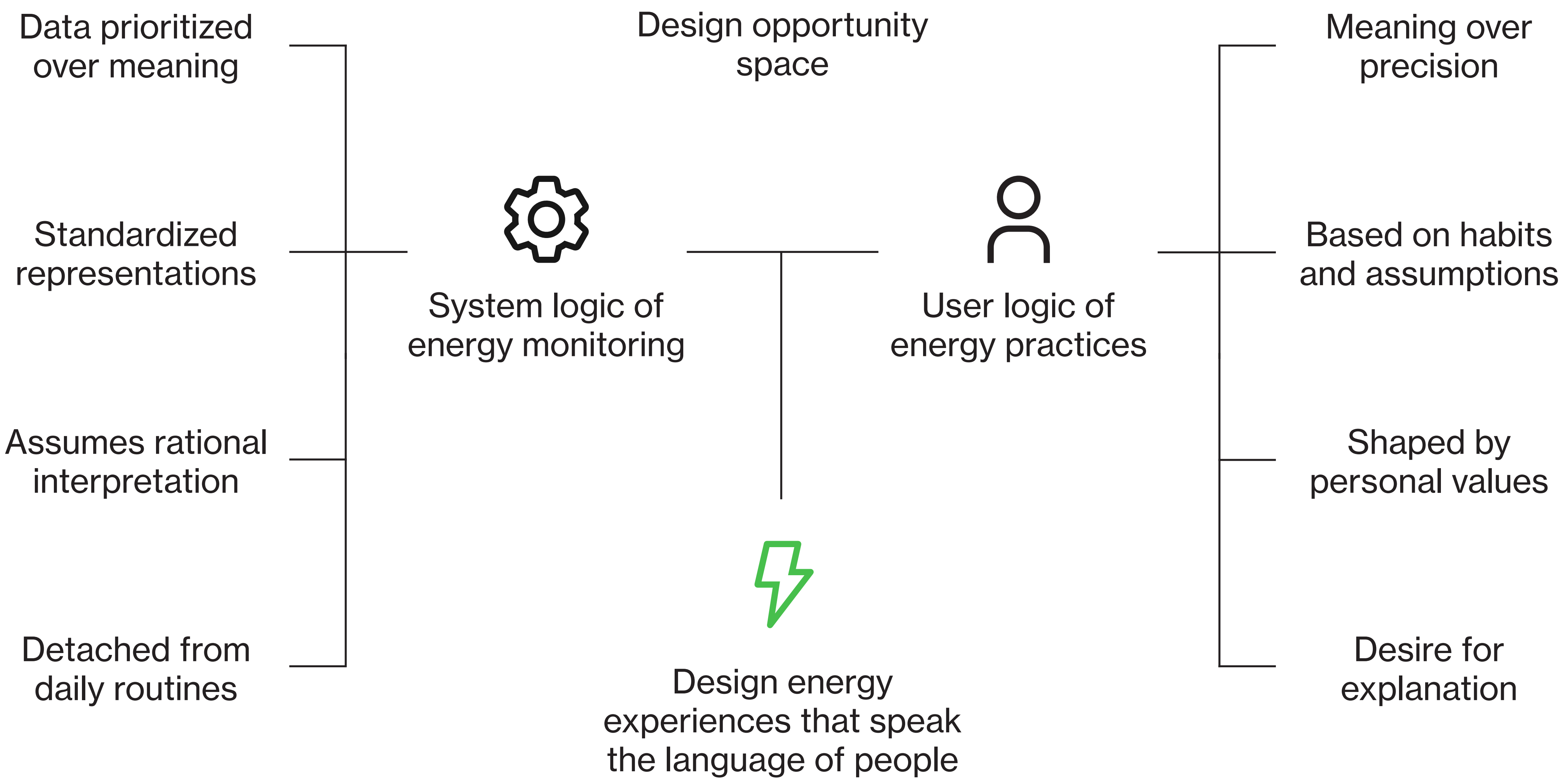
The project was grounded in qualitative user research that explored how energy consumption is experienced and interpreted in everyday life.

We conducted interviews with participants living in a wide range of household situations within the Finnish context, from students in shared apartments to families with children and professional couples. Despite this variety of living contexts and life stages, research revealed a strong overlap in motivations and behaviors.

Insights were synthesized into user personas capturing recurring mental models and habits around energy consumption, framing energy consumption as a situational experience shaped by everyday routines.



From monitoring to sense-making



Research confirmed the assumption that existing energy apps are often experienced as passive dashboards. They show data that are easy to forget and not in context.

Participants expressed uncertainty about the cause-effect relationship between everyday actions and energy consumption, often relying on assumptions or inherited habits rather than clear understanding. At the same time, many already had personal logics for saving energy, even when these did not significantly affect costs.

The research also highlighted clear limits to the role of AI in this context. AI was welcomed as a guide that explains and contextualizes energy use, while control and automation were perceived as intrusive.

Building on these insights, EnergyLM is guided by a set of design principles that translate research findings directly into interaction and system-level decisions.

EnergyLM treats interpretation as the core interface, favoring narration and explanation over raw charts so energy data is always meaningful and contextual. Awareness is framed through everyday situations and routines, allowing users to clearly understand cause-effect relationships within their own lives.

The system builds on existing user logic and habits, supporting reflection without overriding personal values. AI is positioned as an interpretive guide, offering explainable and reversible suggestions while preserving user agency and transparency.

Interpretative by Default

Explain energy data through meaningful narratives

Activity Understanding

Machine learning groups energy use by activities

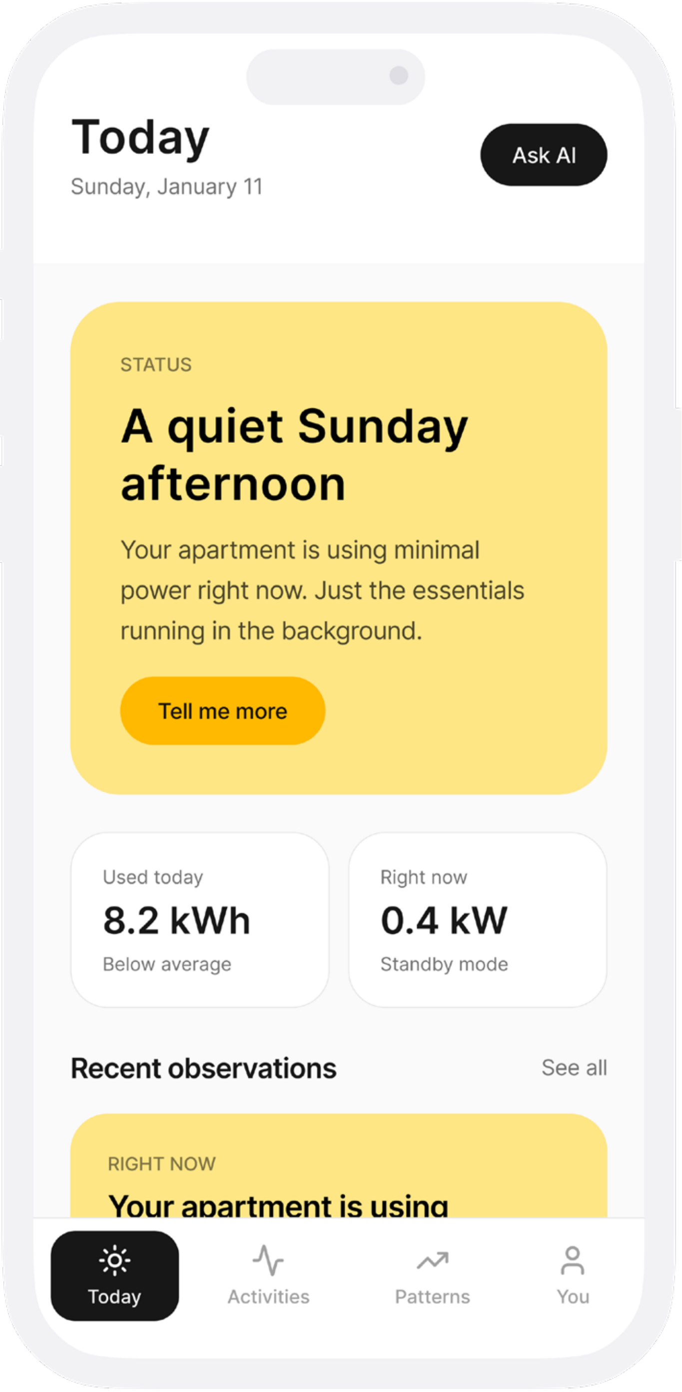
EnergyLM

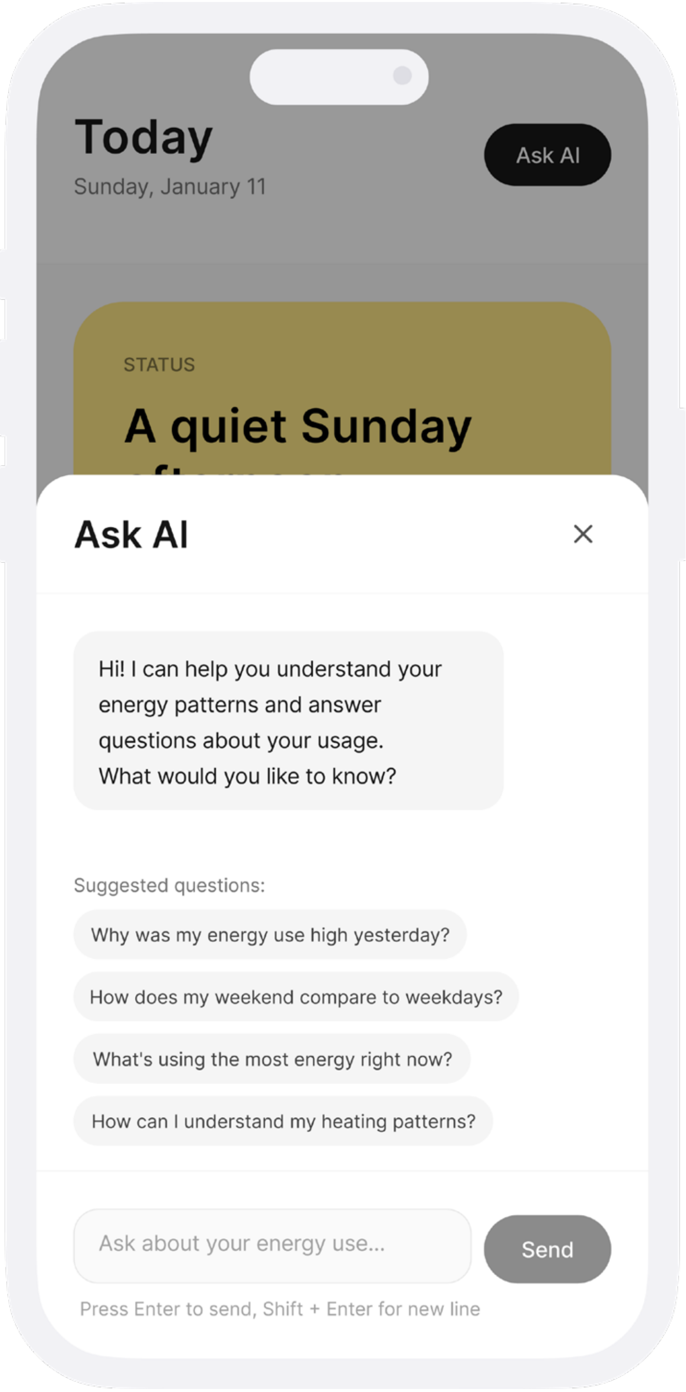
User-Provided Context

Users add context, ask questions, and reflect

Safe and Bounded AI

AI interprets given data, never automates optimization



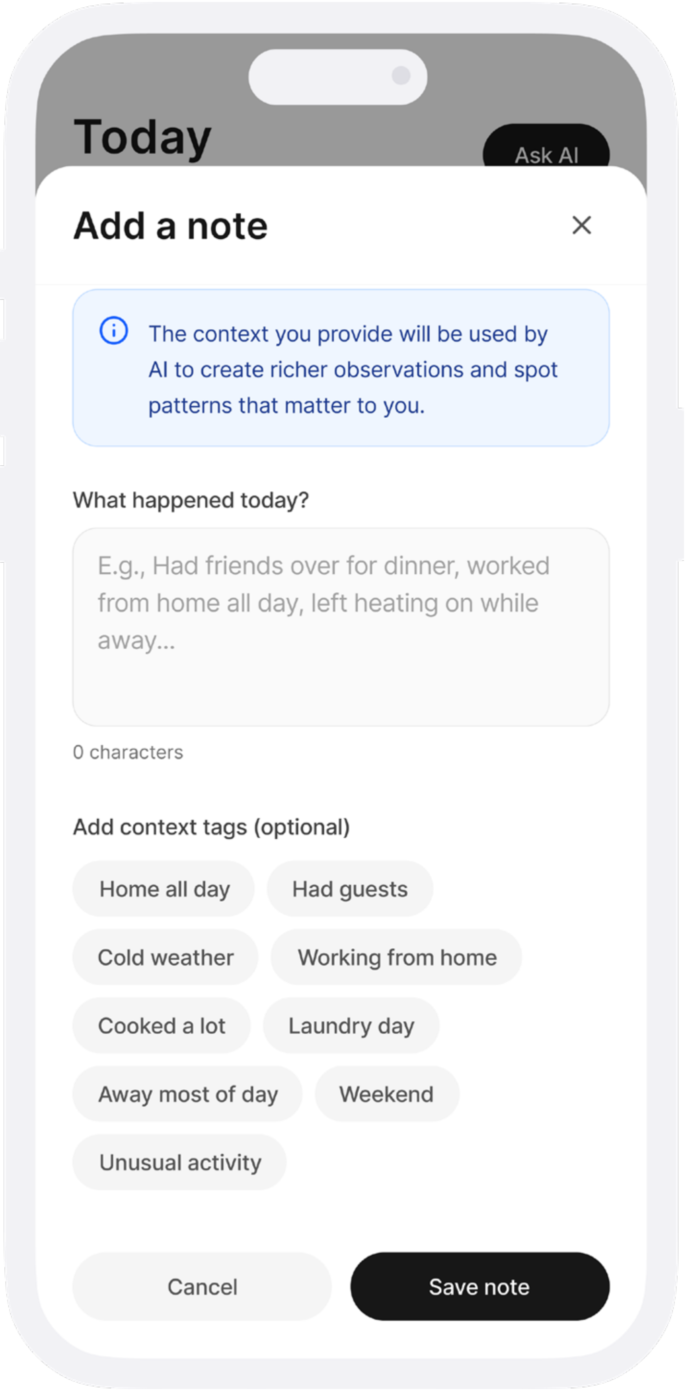


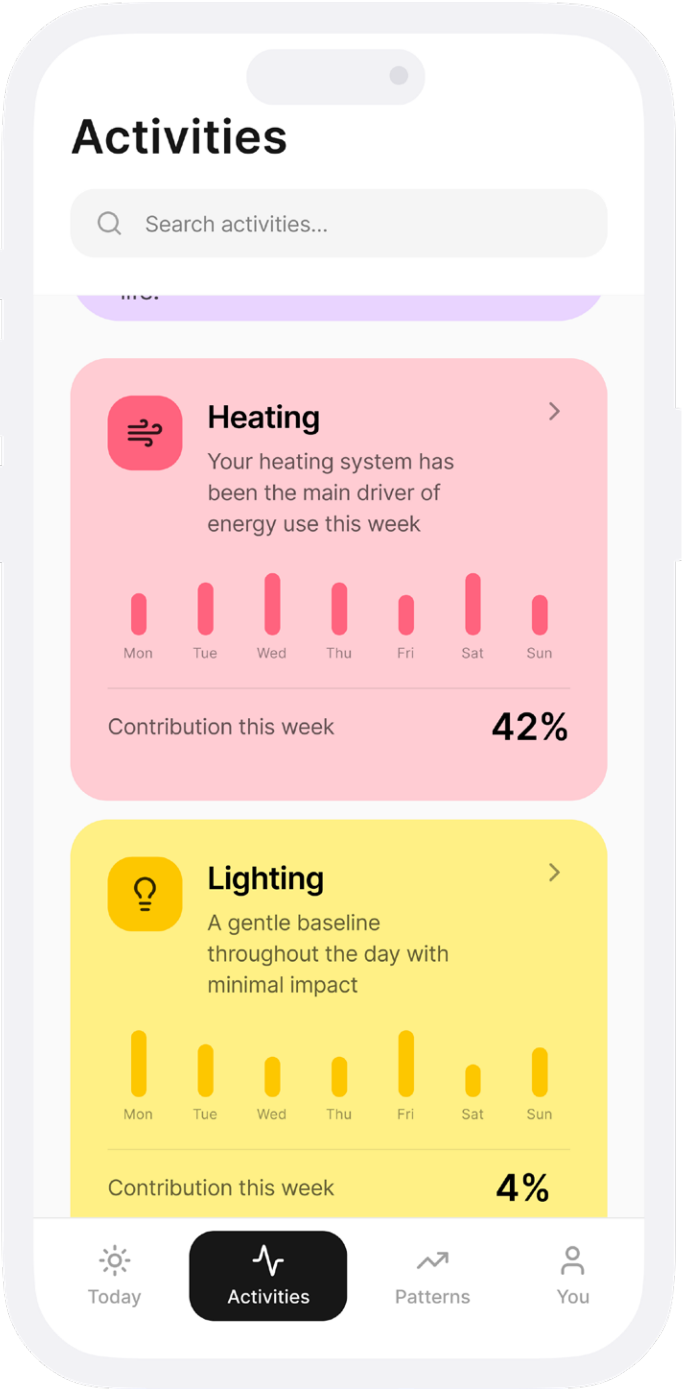
Today

Summarizes the current situation using recent data and observations, helping you quickly understand what is happening right now and why it looks the way it does.

Ask and Notes

Ask questions or add notes about daily activities and situations, allowing the AI to interpret usage patterns through real-life context instead of raw numbers alone.



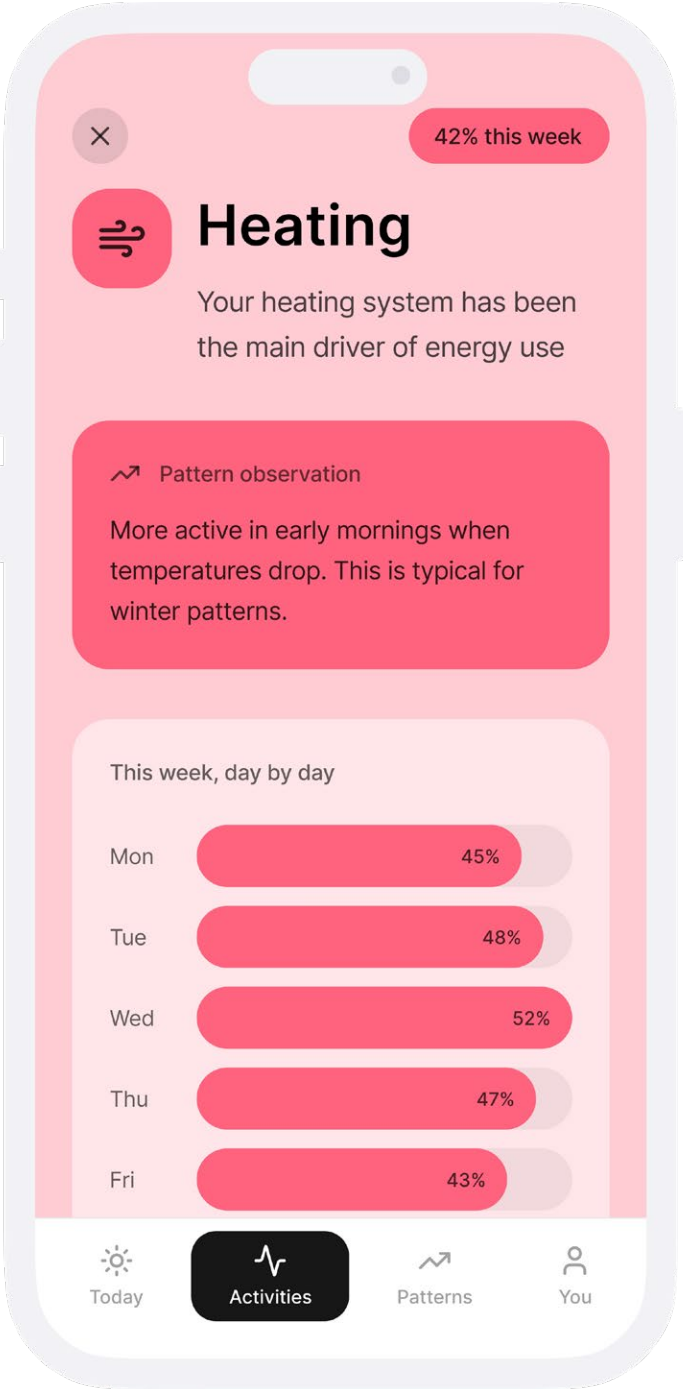


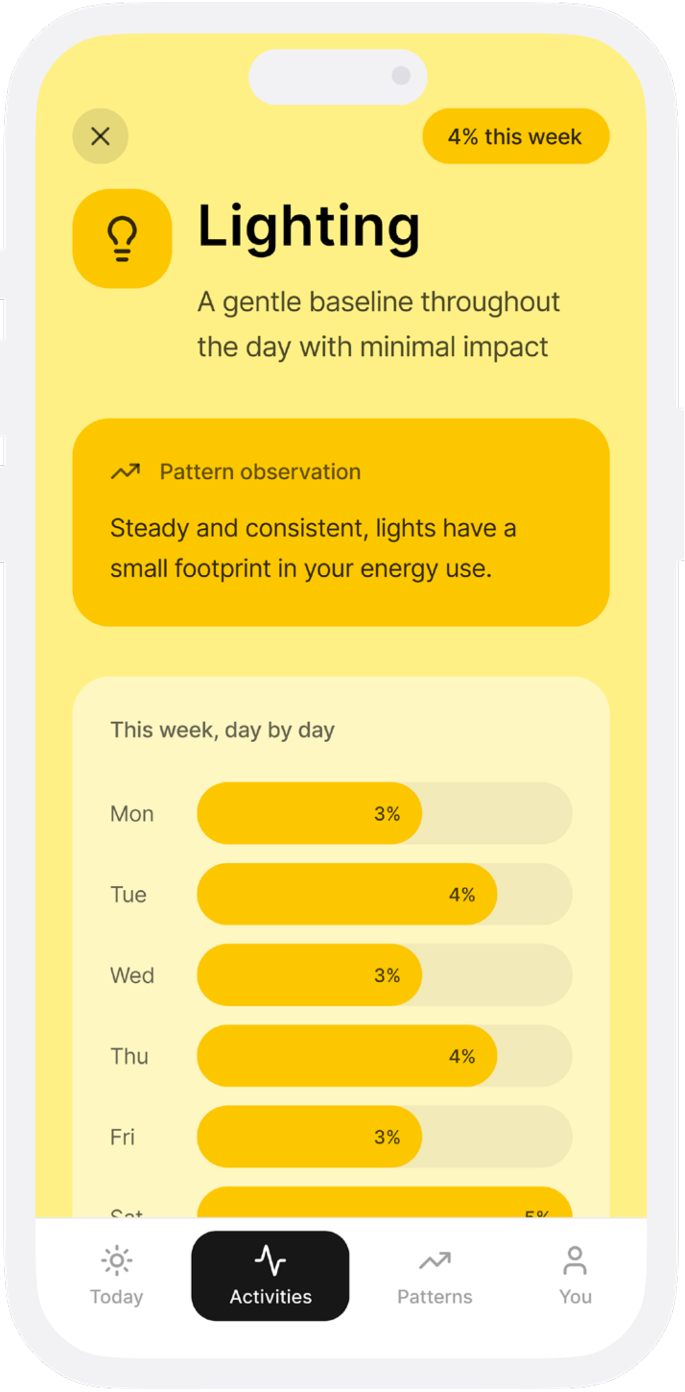
Energy Categories

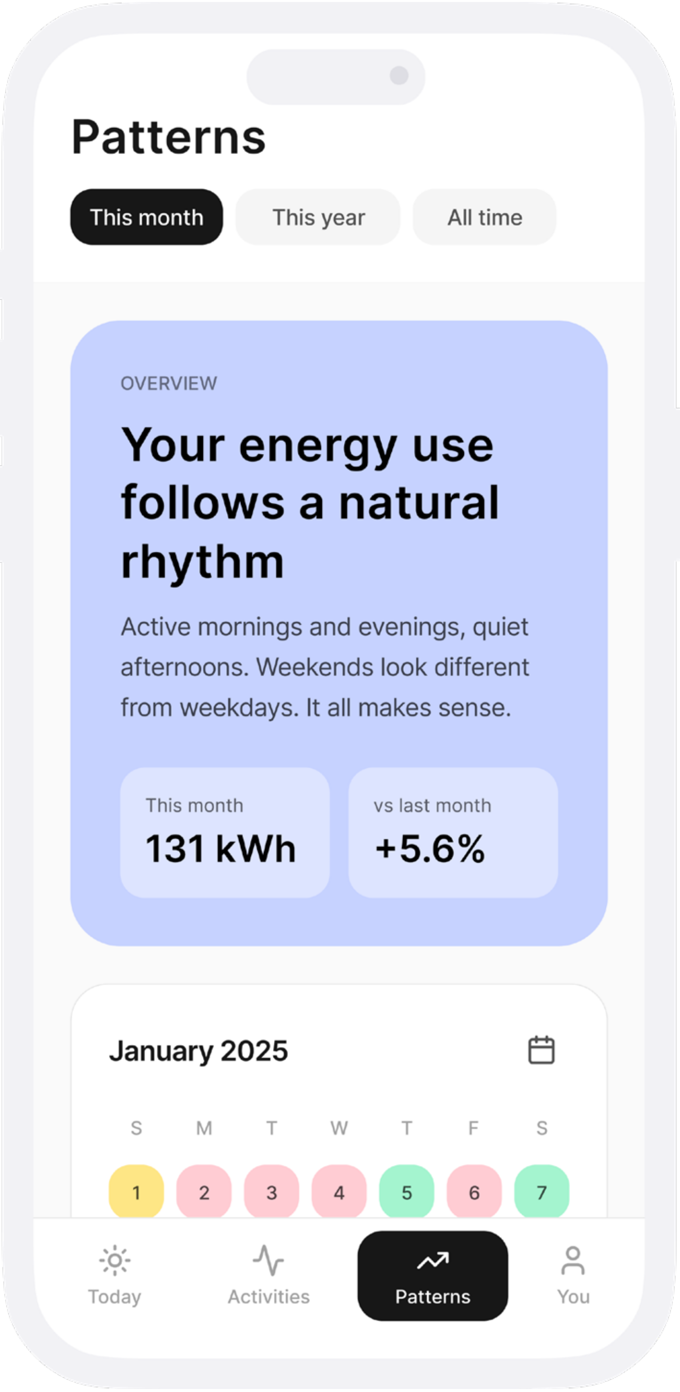
Uses machine learning to recognize and group energy use into meaningful activity categories, making complex consumption data easier to read and compare.

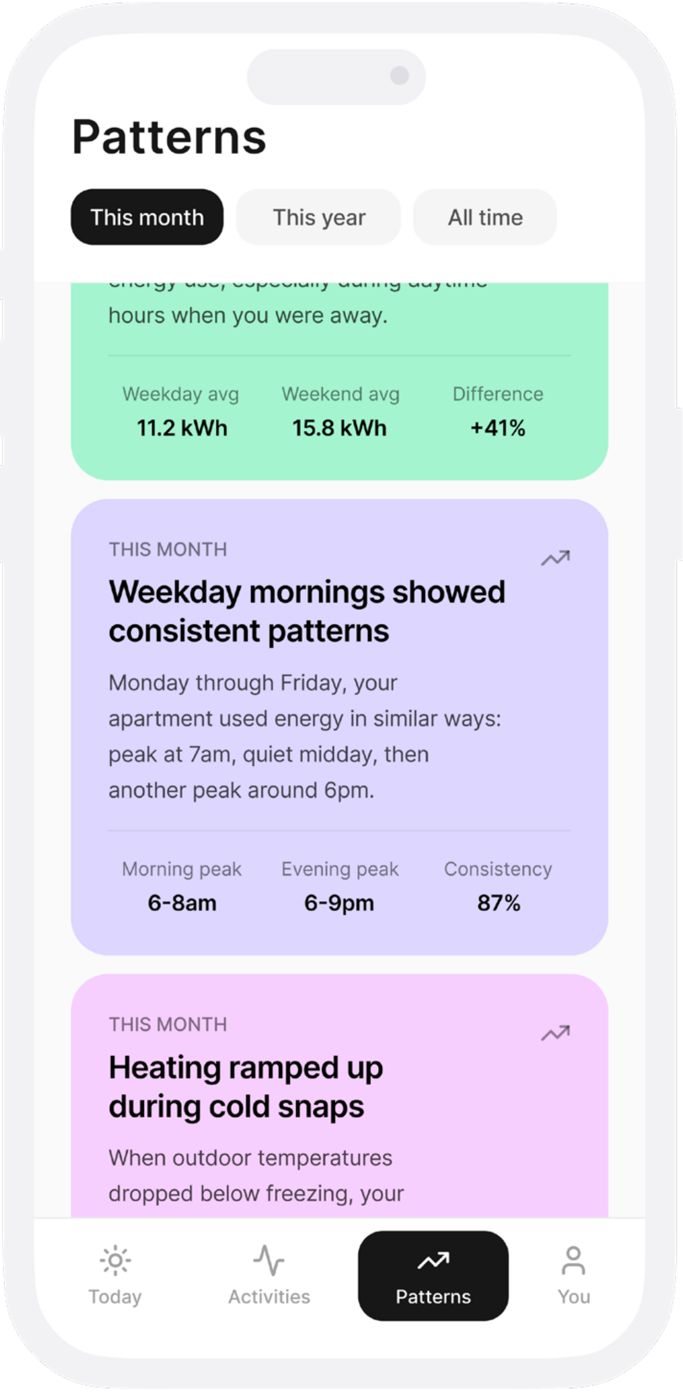
Activity Details

Provides a detailed view over time, highlighting usage trends and linking them to contextual factors such as routines, notes, or external conditions.







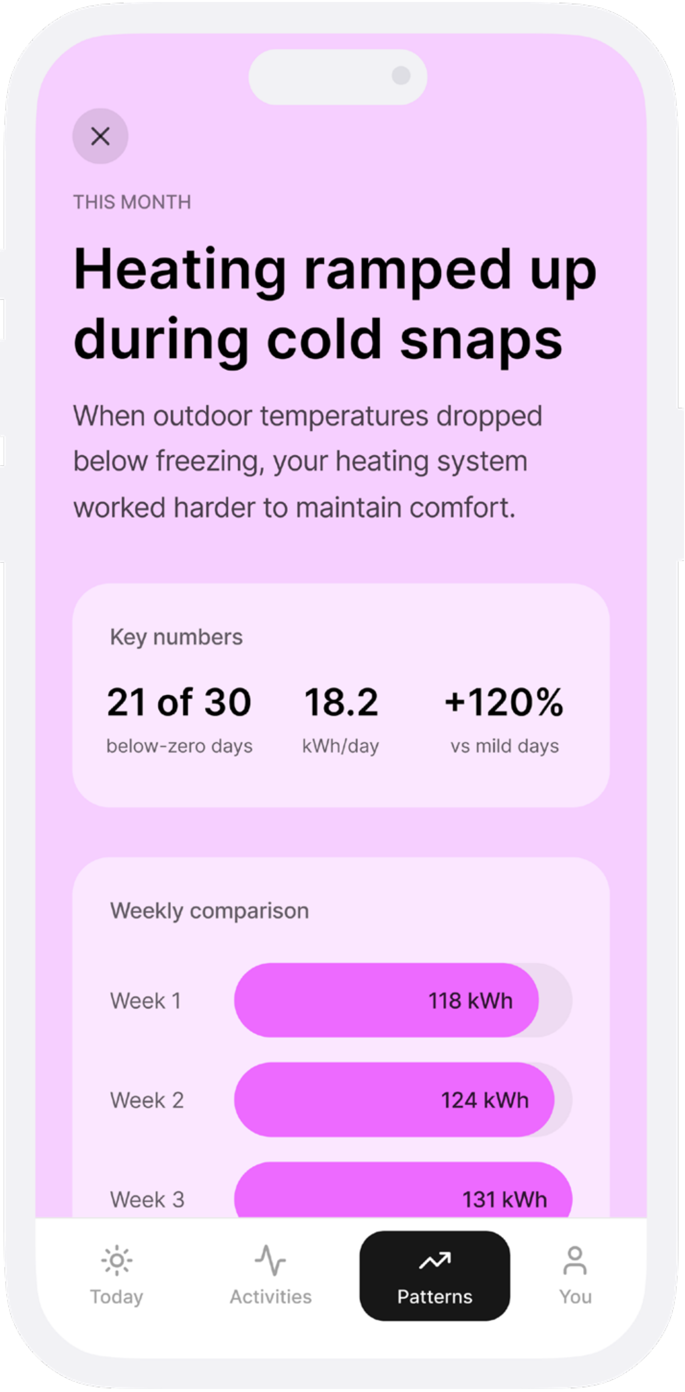


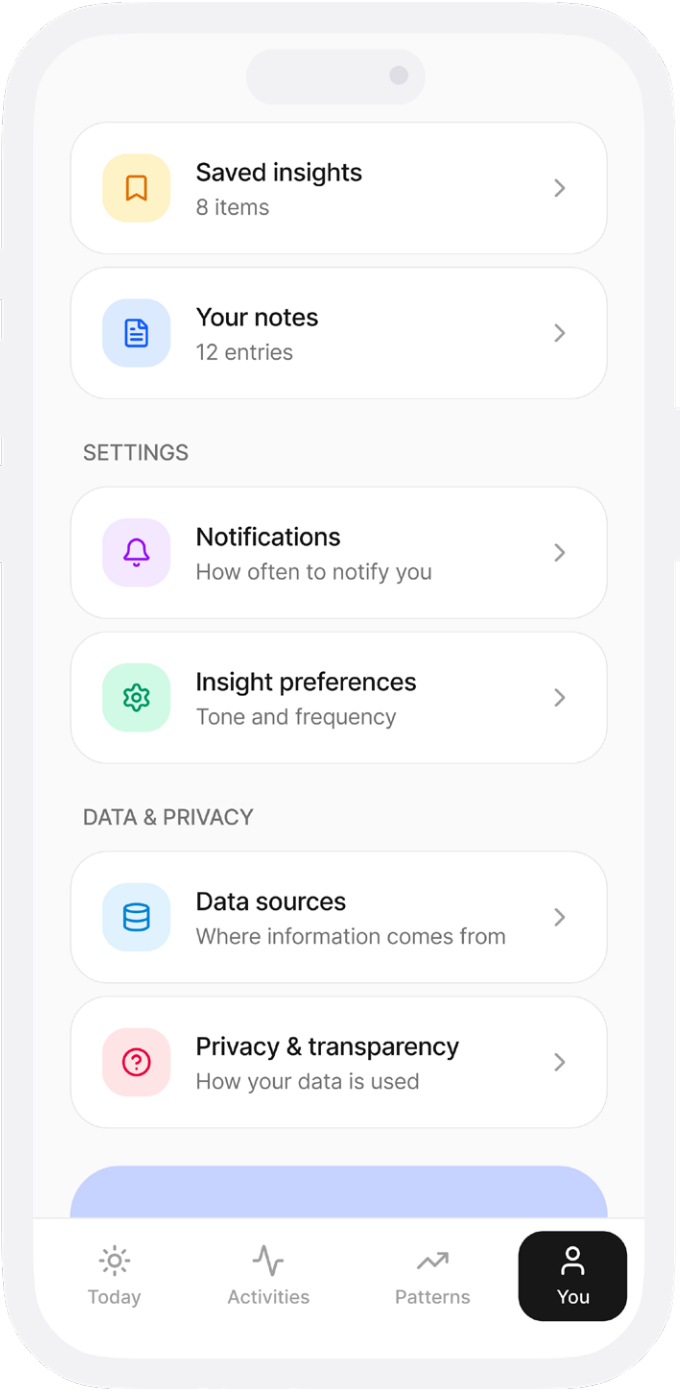
Recognize Trends

Identifies recurring behaviors and shifts over time through a clear calendar-based view, turning long-term data into understandable trends.

Pattern Insights

Interprets key metrics behind each pattern, helping you reflect on causes and empowering informed, conscious decisions rather than reactive actions.





Full Control

Review, edit, or remove notes and data sources, and choose which contextual information the AI can access, ensuring transparency and personal control.

Saved Insights

Save meaningful observations and reflections to revisit later, building a personal archive of insights that supports long-term awareness and learning.